Zabir Saleh

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CAREER SUMMARY

Skilled RAN & RF Engineer with expertise in GSM, WCDMA, LTE & 5G NR network solution, integration and operational activities. Hands-on experience in multi-vendor network and multi access technology. Proven track record includes working with accuracy and speed on large scale network transformation projects e.g., 5G-NR Trial & Commercial launch, VoLTE, MIMO, MBB evolution, 3G and LTE Deployment, massive roll-out, swap and carrier merging. I want to pursue my career in fast-paced IT, Telecom or Engineering industry where I can contribute my knowledge, skill & experience as well as enrich myself with continuous evolution of cutting-edge technologies.

EXPERIENCE SUMMARY

OEM Vendor Hands-On: Ericsson, Nokia, Huawei [Ericsson RF & RAN Expert – 10 Years]
Experience in Telecom Operators: STC, Mobily [KSA], Etisalat [UAE], T-Mobile, US Cellular [USA], Vodafone [Spain, Portugal, UK], EE-BT [UK], MCI [Iran], Telefonica [Brazil], Axiata [Bangladesh, Malaysia], DNB [Malaysia]

PRODUCT HANDS-ON

RAN, Core, IP-RAN Transport & Transmission Product end-to-end hands on:

- ERICSSON: BB5216, 6630, 6648, 6631, RP 6651, 6655, 6672
- ERICSSON: DUS/DUL, DUW, DUG, DXU, CBU, RBS6000, 2000
- ERICSSON: SIU, TCU, ROUTER 6000, MINI-LINK TN
- ERICSSON: BSC EVO8200, RNC EVO8200, NWI-E, APZ, APG43L
- ERICSSON: MSC, SGSN-MME, EPC & IMS [GGSN, HLR, VLR, SIP]
- ERICSSON: OSS-RC, ENM, ENIQ, BO
- HUAWEI: BBU3900,3910,5900, BSC6900/6910
- HUAWEI: SGSN USN, MGW UMG8900
- NOKIA: AirScale, Flexi, MR10
- NOKIA: Flexi BSC, mcBSC

RAN & RF Planning Hands-On Tools:

- RF Planning: Atoll, Atoll One, Planet, ASSET Radio
- RF Optimization: TEMS Investigation, Discovery, Pocket; XCAL, XCAP; NEMO Outdoor, Analyze
- RAN Performance: ENM BO, ENIQ, NetACT, Huawei iMaster MAE,
- Microwave: Pathloss, Infovista Ellipse
- GIS: MapInfo, ArcGIS

WORK EXPERIENCE DETAILS

RF & RAN CONSULTANT – CELFINET PT - CYIENT

Name of The Company: CELFINET PT
Department: 5G RF & RAN Support – DNB Malaysia
Expertise: 2G, 3G, LTE, 5G RAN
Position Held: RF & RAN Consultant
Duration of Job: April 2022 to March 2026

Job Milestones:

Employed as a RAN engineer and I have been involved with DNB Malaysia's RF & RAN solutions. Followings are my key responsibilities in different projects.

Major Highlights: An active functional technical member of Europe & UK LTE, 5G-NR projects.

- **RF & RAN Solution:** 5G & LTE site solution as per geo and capacity
- **RAN Architecture:** Allocation and distribute BaseBand, RP and Radio Units as per needed
- **RAN 2nd & 3rd Level Back-Office Support:** Support whole Ericsson RAN whole network on integration, architecture, design & optimization troubleshooting
- **5G NR Integration Activity:** End-to-End configuration & pre-on-air of newly deployed sites
- **LTE Modernization:** Solution integration activity of LTE_700, LTE_1800, LTE_2100 & LTE_2300
- **5G Pre-Launch Verification:** Integration test, feature verification activity 5G anchoring with eNodeB
- **ENM & OSS Migration Activity:** ENM Configuration verification activity for 5G, 4G, 3G Nodes
- **Ericsson Dynamic Spectrum Trial:** Testing & Performing trials on dynamic spectrum trials & analysis on 4G & 5G
- **MPLS & IP Network Task:** IP-RAN modernization from STN L3 switch TCU/SIU to Ericsson Router 6000 Series
- **Sync Solution:** Solution of different syncs such as PTP, GNSS, E-RAN, NTP, Sync_E
- **Project & O&M CSR:** Handling network faults with ELS, PDU directly
- **SW Upgrade:** Perform & Managing annual SW upgrade of RAN nodes [BSC, RNC, DU, BaseBand, RAN Processors]

SENIOR RAN ENGINEER – ERICSSON AB

Name of The Company: Ericsson AB (Saudi Arabia)
Department: RAN Support
Expertise: 2G, 3G, LTE, 5G RAN
Position Held: RF & RAN Engineer
Duration of Job: From July 2017 to March 2022

Major Highlights: 5G launch in KSA; Integration & Commissioning of 3 EVO BSCs, 5 RNCs, 1 ENM.

Job Milestones:

Employed as a RAN support, integration, back-office engineer. Followings are my key responsibilities in different projects in various parts of the world.

- **RAN 2nd & 3rd Level Back-Office Support:** Support whole STC Ericsson RAN whole network on 24/7 roster-based shifts
- **STC 5G NR Integration Activity:** End-to-End configuration & pre-on-air of newly deployed sites
- **STC LTE Modernization:** Solution integration activity of LTE_700, LTE_1800, LTE_2100 & LTE_2300
- **5G Pre-Launch Verification:** Integration test, feature verification activity 5G anchoring with eNodeB
- **ENM & OSS Migration Activity:** ENM Configuration verification activity for 5G, 4G, 3G Nodes with Integration engineers
- **Ericsson Dynamic Spectrum Trial:** Testing & Performing trials on dynamic spectrum trials & analysis on 4G & 5G
- **MPLS & IP Network Task:** IP-RAN modernization from STN L3 switch TCU/SIU to Ericsson Router 6000 Series
- **Sync Solution:** Solution of different syncs such as PTP, GNSS, E-RAN, NTP, Sync_E
- **Project & O&M CSR:** Handling network faults with ELS, PDU directly
- **SW Upgrade:** Perform & Managing annual SW upgrade of RAN nodes [BSC, RNC, DU, BaseBand]
- **5G Trial:** An active member of 5G RAN trial team to support field team during implementation & commissioning of BaseBand & AIR radio units.
- **L_700 & L_1400 Project:** Need to handle CRs related to that project to take care of any integration & critical hiccups within SLA
- **Antenna Modernization:** Existing Single, dual, tri & penta band antennas are modernized by new octa-band antenna.
- **STC Modernization Project:** Support integration, field, deployment, FO, BO during the CRs to smoothen up the delivery & execution of the project within the SLA.
- **SW Upgradation:** BSC, RNC, eNodeB, NodeB, BTS soft upgradation activity with CSR & Integration team
- **IP Transport:** Support integration & roll-out activity of newly added SIU, TCU & MW network
- **LTE TDD & FDD Projects:** Integration activity of U900, U2100, FDD_1800 & FDD_2100, TDD_2300 & TDD_2600
- **Day-to-Day O&M Activities:** Back-Office & Integration support to front office & field team for any maintenance trouble tickets, PATs, CRs

RAN BO ENGINEER - AXIATA LTD

Name of the Company: Axiata Ltd (Bangladesh)
Department: Central Operations [Back-Office]
Expertise: RAN, Core & Tx Integration, Configuration & Operation
Position Held: RAN BO Engineer [Integration & Configuration]
Duration of Job: January 2014 to June 2017

Major Highlights: First E/// BaseBand integration in Robi in-house, handled all over 3G on-air in Bangladesh

Job Milestones:

Employed as a RAN & Core BO integration engineer for nearly 3 and half years. I have been involved with major turn-key revenue generated projects of Axiata Bangladesh. I have been exposed in multi-vendor environment and multi domain environment. Followings are my key responsibilities in different projects.

- **Axiata 3G Deployment:** Integration of nearly 8000 3G 2100 MHz sites in whole network. Integration of newly deployed Ericsson DUW, Ericsson BaseBand, Huawei BBU3900, BTS3900_U, Nokia MR10
- **Axiata GSM Falcon Project:** Integration of new 1000 2G 900 & 1800 MHz sites for coverage extension purpose. Newly deployed nodes are mainly E/// DUGs, also major Huawei BTS & Nokia Flexi & MR10

- **Axiata Nokia-to-Ericsson Swap:** Integration of deployed Ericsson RBS and swapping Nokia in a whole cluster of 5000 sites in key Comilla & Chittagong zone.
- **Axiata Airtel Merger:** This is a unique network merging project. Integration involved in every domain such as RAN, Core & Transmission. We have to merge the whole Core & RAN network of two operators in a complex multi-vendor environment. It involves around 12 thousand live sites, 100 BSCs & RNCs, 12 MSC, 6 SGSN-MME
- **Axiata LTE Trial Project:** Integration of around 50 pilot LTE sites as trial prior to the commercial on-air of 4G LTE in whole network.

TECHNICAL SKILLS & DOMAIN COMPETENCIES

As a solution integration, L2 & L3 support, and Back-Office Engineer for more than 12 years, I have gained extensive hands-on experiences & been tasked with comprehensive responsibilities in RAN, Core, IP-RAN Transport & Transmission domains. During that period, followings are my job responsibilities:

RAN & RF Domain Skills:

- E2E services related Hands-on Multi-Vendor (Ericsson, Huawei, Nokia) gNodeB, eNodeB, BSC, RNC RAN Nodes
- Commercial Integration, acceptance and troubleshooting acceptance of 5G BaseBand & AIR Units.
- 5G NR NSA/SA RAN Access Product Integration
- Troubleshooting of wireless system radio interface L2 / L3 protocols (RRC / SDAP / PDCP / RLC / MAC), network & system architecture and functionalities
- 5G Hands on design and optimization implementation (NSA, SA, DSS, Signaling flows, Mobility, low and mid band, MMWave, DL/UL split bearer, ENDC resource utilization)
- 5G NR NSA SW Basic Verification (Regression Test) & Feature Verification both in Lab and Live
- 5G NR NSA Issue Identification, CSR/TR writing
- 5G & Anchoring-LTE issues resolution with PLM Support; and PDU for issue verification and product development. DCGM, ESI and EMCA log analysis)
- Using XCAL tools for testing QoS, QoE
- Live traffic log analysis with TEMS tool for 3GPP signaling message decoding and analysis
- Packet Tracings analysis of different interfaces with Wireshark
- Calibrate the performance of network service (Voice, FTP, HTTP, Multi-RAB, etc)
- EPC signaling tracing & testing for 5G nodes
- 5G NR anchoring with LTE node interface issue rectification
- Instant/Dynamic Spectrum Sharing LTE - 5G NR ESS (Ericsson Spectrum Sharing) Test, Trial, Commercial Integration
- 5G NR NSA High-Band/mmWave + Mid-Band/Sub-6 GHz indoor (IBS) and outdoor (macro) Integration
- HLD, LLD acceptance & verification for 5G sites integration in E/// ENM Server
- 5G Mass Roll-Out support and scripting through MSDP PCI Tools
- Implementation of AP (Auto Provisioning) & AI-ZT (Auto Integration Zero Touch) of 5G RAN Nodes
- Continuous 5G features testing on various interface and IP sctp-association issues
- FNI of ORAN, Cloud RAN, 5G-NR Evolution Path, ML & AI applicable to RAN services
- Analyze technology requirements to support technical based on Ericsson's ML/AI based cognitive services portfolio
- Network Sync related interface issue troubleshooting with Router 6672
- Ericsson OSS-RC to ENM migration node related activities for different domain
- Commissioning & start-up Integration of EVO BSC 8200, EVO RNC 8200, APG43L, APZ 21255, APZ 21260
- Remote integration of E/// DUG, DUW, DUL, DUS, BaseBand, PICO, RDS
- Performing different re-homing and re-parenting activity of 2G, 3G, BSC, RNC
- Perform SW upgradation of different E/// RAN nodes – DUG, DUS, BaseBand

- Perform APG43L SW upgradation through OSS-RC, ENM or manual innervation
- Perform routine health check of EPB, CMXB and SCXB boards of EVO 8200 RNC
- 3rd Level tracings of BSC MOs, UETR, CTR, RNC GPEH, LTE CTR to determine faults
- Have great understanding of 3GPP messages in call flow to determine and support RF teams to address different architecture and NW level issues
- Handling major RF different issues from L3 support perspective – VoLTE & CSFB degradation in LTE, CS/PS Drop in BSC/RNC
- Working with L2, solution architects and PDU R&D to overcome and finalize roll-out and operational issues
- QoS related trials and tracings decoding
- Decode different tracings files
- Automate roll-out scripts for fast integration and operational activities with python & bash scripts
- Hands-on experience on TEMS, INFOVISTA, PLANET, ASTELLIA, PATHLOSS for different RF, CCD, Microwave, RAN-Transport, IP MPLS network
- NodeB OAM & IuB interface IP, VLAN configuration in SIU & TCU during roll-out and re-homing
- Perform software upgradation of DUW through SMO
- Remote level integration execution and support on migration from DUW20, DUW30 to DUW31, DUW41
- Perform routine health check of EPB Blades, CMXB and SCXB boards of EVO 8200 RNC
- Integrate different interfaces of RNCs with other core & RAN nodes as RNC, MGW, SGSN nodes e.g. IuR, IuCS, IuPS interfaces
- Install LKF file of NodeB through WinFlol & RBS EM as well as in bulk through SMO
- Prepare & configuration of re-homing of RNC from one MGW, SGSN to another MGW, SGSN

CS & PS Core Domain Skills:

- Integration of RAN parameters e.g. LAC, RAC, TAC, CGI in MSC, SGSN, MME
- Day-to-Day RAN traffic analysis in CS & PS Core from OSS
- Configuration of signaling & voice channel of TDM and IP signaling
- Various interface resource integration (A/E, Gb etc)
- Interconnect integration & configuration with core nodes (GMSCs)
- VoLTE functions & CSFB parameter configuration, testing & trial with EPC
- Integrate MGW with MSC server, RNC, BSC, STP, other MSC, PSTN switch, IN/VAS, HLR
- Configure Traditional C7 Signaling and SIGTRAN and ATM
- Nb over IP (Jitter, Remote IP, UDP host, Access host, IP interface and Gigabit interface) and BICC (SAE and AXE Parameter)
- MGW/RNC statistics/PM like Jitter, Remote IP, NB calls, ATM calls, C7 signaling, ATM signaling and SIGTRAN

Tx & IP Transport Domain Skills:

- Integration and configuration of SIU, TCU, MiniLink-TN
- Installation and configuration of 1+0, 1+1, 1+SD, 2 +0 and XPIC links
- Configuration of NE, Packet Link, Radio Link and VLANs
- IP configuration & dimensioning of NodeB, eNB, SIU, TCU as per ATND low level design
- Abis level IP, VLAN configuration in SIU and TCU during new BTS roll-out and BTS re-homing
- NodeB OAM & IuB interface IP, VLAN configuration in SIU & TCU during roll-out and re-homing
- eNodeB OAM & S1 interface IP, VLAN configuration in SIU & TCU during roll-out and re-homing

RF & RAN Tools Hands-On:

- **Ericsson NMS & OSS Tools:** Ericsson OSS-RC, Common Explorer, WinFiol, OSS Explorer, Moshell, Software Management Organizer, Software Hardware Manager, CNA, BSIM, Bulk CM, Re-Parenting Manager, RBS Element Manager, SOEM, RBS Local OMT, Remote OMT, Mini-Link TN Craft, RNO, NEPA, EXERT,

- **Huawei NMS Tools:** M2000, U2000, WebLMT, Local_LMT, RTN_Web LMT, CME, Genex Probe & Assistant, UNet
- **Nokia NMS & OSS Tools:** Nokia Telnet & SSH 'TANG' Tool, Nokia NetAct, Nokia BTS Manager
- **RF Tools:** TEMS Investigation & Discovery, ACTIX ONE & Analyzer, MapInfo, XCAL, XCAP,
- **Misc. Network Tools:** Putty, TelNet, Cygwin, SecureCRT, TeraTerm, GNS3, Cisco Packet-Tracer, WireShark
- **Telecom Applications & Tools:** PLANET, BMC Remedy, CRM Tool, STC WFMS, Axiata WO Tool
- **Tools:** CodeBLock, MS Office Automation Tool
- **Coding:** MML, XML, shell Scripting, C, Python, Visual Basic Programming for Script Automation

EDUCATION

Degree:

Degree: MSc in Computer Science
Institution: University of Technology Malaysia (UTM), Year: 2027

Degree: BSc in Electrical Engineering
Institution: Ahsanullah University of Sci & Tech, Year: 2013

High School:

Institution: Bangladesh International School & College, Riyadh, Saudi Arabia
GCE A-Level, Year: 2008
GCE O-Level, Year: 2006

ACADEMIC PROJECTS

RAN Digital Twin – Axiata (FYP)
Tech: Python, 4G/LTE, 5G NR, Digital Twin, RAN Simulation Analytics Analysis, Pandas, NumPy, Scikit-learn

FINTRUST – Zero Trust FinTech & Blockchain Application
Tech: Flutter, Dart, Node.js, Express.js, MySQL, Firebase, REST API, MFA, OTP, SHA-256, Ethereum Sepolia

Incident Hub – Incident Management & Automation Platform
Tech: React, JavaScript, Node.js, Express.js, MySQL, REST API, JWT, UiPath, Git

GreenLeaf – Sustainability Management Web Application
Techs: Java, Spring Boot, Spring MVC, Thymeleaf, MySQL, HTML, CSS, JavaScript, Maven

LazyWave – AI-Driven 5G RAN Network Intelligence Platform
Tech: Python, Scikit-learn, Pandas, NumPy, Random Forest, SVM, Neural Networks, 5G NR

PROFESSIONAL TRAININGS AND CERTIFICATIONS

- Cisco Certified Network Associate [CCNA]
- Ericsson BaseBand Comprehensive [2G, 3G, 4G, 5G]
- Ericsson RAN RBS 6000 BTS, NodeB, eNodeB
- Ericsson NB-IoT & 5G evolution

- Ericsson RAN RNC EVO-C 8200 Operation, Configuration
- Ericsson RAN BaseBand Nodes
- Ericsson GSM RAN BSC EVO Controller 8200 BSC Operation, Configuration
- Ericsson GSM Basics
- Huawei BBU 3900 Product Configuration & Troubleshooting
- Huawei BSC/RNC 6900/6910 Commissioning
- Huawei BTS 3900 gBTS, eBTS, uBTS
- Huawei CS Core Fundamentals [MSOFTx]
- Huawei PS Core Fundamentals [SGSN USN Nodes]
- Nokia MR10 & Flexi BTS Configuration
- Nokia Flexi & mcBSC
- Nokia IP-RAN Juniper Module